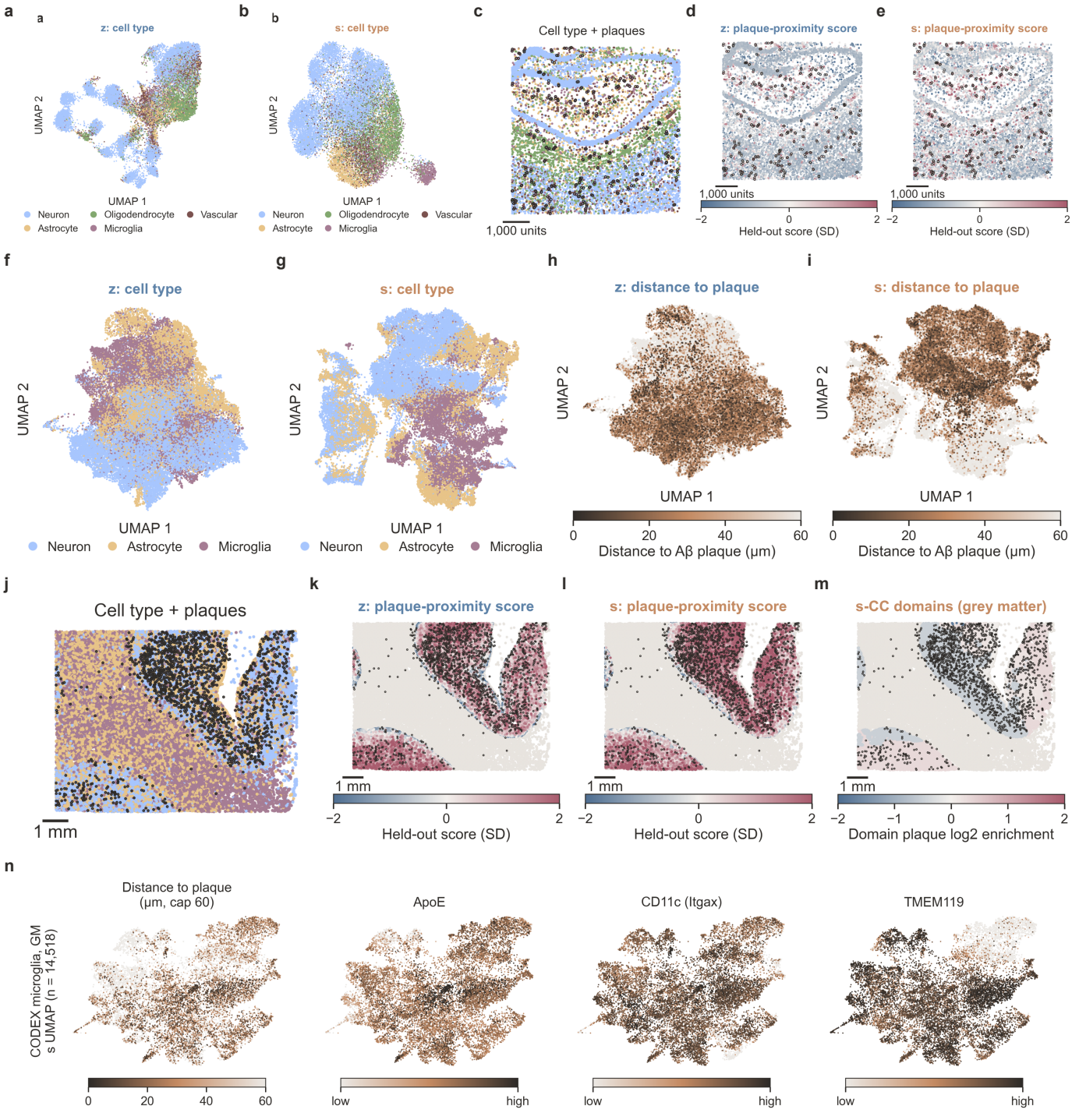

a
z: cell type
UMAP 2
UMAP 1
Neuron
Oligodendrocyte
Vascular
Astrocyte
Microglia
b
b
s: cell type
UMAP 2
UMAP 1
Neuron
Oligodendrocyte
Vascular
Astrocyte
Microglia
c
Cell type + plaques
1,000 units
d
z: plaque-proximity score
1,000 units
−2
0
2
Held-out score (SD)
e
s: plaque-proximity score
1,000 units
−2
0
2
Held-out score (SD)
f
z: cell type
UMAP 2
UMAP 1
Neuron
Astrocyte
Microglia
g
s: cell type
UMAP 2
UMAP 1
Neuron
Astrocyte
Microglia
h
z: distance to plaque
UMAP 2
UMAP 1
0
20
40
60
Distance to Aβ plaque (μm)
i
s: distance to plaque
UMAP 2
UMAP 1
0
20
40
60
Distance to Aβ plaque (μm)
j
Cell type + plaques
1 mm
k
z: plaque-proximity score
1 mm
−2
0
2
Held-out score (SD)
l
s: plaque-proximity score
1 mm
−2
0
2
Held-out score (SD)
m
s-CC domains (grey matter)
1 mm
−2
−1
0
1
2
Domain plaque log2 enrichment
n
Distance to plaque
(μm, cap 60)
0
20
40
60
ApoE
low
high
CD11c (Itgax)
low
high
TMEM119
low
high
CODEX microglia, GM
s UMAP (n = 14,518)